

# TECHNICAL BRIEF: ConvNeXt-CBAM FOR BREAST ULTRASOUND

Evaluating Lightweight Dual-Attention Architectures for Clinically Aligned & Explainable Classification

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**Focus Area:** Clinical Readiness, Parameter Efficiency, and XAI Visual Verification

## 1. Executive Summary & Objective

Deep learning frameworks in medical image analysis often face two core hurdles in real-world deployment: high computational footprints that hinder edge deployment on clinical hardware, and a lack of interpretable feature maps aligned with standard diagnostic criteria (e.g., BI-RADS margins).

This project demonstrates a lightweight proof-of-concept pipeline utilizing a modernized convolutional backbone (**ConvNeXt-Tiny**) augmented with a **Convolutional Block Attention Module (CBAM)**. Evaluated on the Dataset of Breast Ultrasound Images (BUSI), the proposed architecture achieves high diagnostic efficacy (**0.93 AUC-ROC**) while retaining low parameter complexity (**28.6M parameters**) and localized feature attribution via Grad-CAM visual explanations.

## 2. Architectural Design

The framework integrates sequential spatial and channel attention mechanisms into the deep feature extraction stages of ConvNeXt-Tiny:

- **Backbone (ConvNeXt-Tiny):** Replaces traditional heavy ResNet/VGG backbones. Utilizes  $7 \times 7$  depthwise separable convolutions to expand the receptive field without increasing parameter count.
- **Channel Attention Module (CAM):** Aggregates spatial information using both Average Pooling and Max Pooling simultaneously. A shared multi-layer perceptron (MLP) calculates channel-wise weights ( $\mathbf{M}_c$ ), prioritizing relevant acoustic feature channels while suppressing background speckle noise.
- **Spatial Attention Module (SAM):** Applies inter-spatial operations across channels, generating a 2D spatial attention map ( $\mathbf{M}_s$ ) via a  $7 \times 7$  convolution. This forces the network to concentrate explicitly on acoustic shadowing, structural distortion, and lesion boundaries.

Mathematical formulation for feature recalibration:

$$\mathbf{F}' = \mathbf{M}_c(\mathbf{F}) \otimes \mathbf{F} \quad (1)$$

$$\mathbf{F}'' = \mathbf{M}_s(\mathbf{F}') \otimes \mathbf{F}' \quad (2)$$

## 3. Experimental Setup & Results

### 3.1 Dataset & Preprocessing

- **Source:** Dataset of Breast Ultrasound Images (BUSI).
- **Filter Protocol:** Automated parsing script to strip ground-truth mask files (`*_mask.png`) from classification input pipelines, preventing data leakage.
- **Partition:** 80% Train, 20% Stratified Validation across 3 target classes: Normal, Benign, and Malignant.
- **Augmentations:** Random horizontal flips, degree rotations ( $\pm 15^\circ$ ), color jittering, and ImageNet normalization.

### 3.2 Quantitative Performance Matrix

Evaluation prioritized macro-averaged clinical metrics (Sensitivity and Specificity) alongside standard accuracy and AUC-ROC.

Table 1: Performance Comparison across Architecture Variants on BUSI Validation Set

| Architecture Variant               | Params        | Train Loss    | Val Loss      | AUC-ROC       | Sensitivity   | Specificity   |
|------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| ResNet-50 (Baseline)               | 25.6 M        | 0.3812        | 0.4120        | 0.8410        | 0.7820        | 0.8210        |
| ConvNeXt-Tiny (Vanilla)            | 28.0 M        | 0.2415        | 0.3150        | 0.8850        | 0.8240        | 0.8650        |
| <b>ConvNeXt-Tiny + CBAM (Ours)</b> | <b>28.6 M</b> | <b>0.1120</b> | <b>0.2105</b> | <b>0.9320</b> | <b>0.8890</b> | <b>0.9210</b> |

#### 4. Explainability & Clinical Visual Alignment

To validate the model from a radiologist-centric perspective, Grad-CAM feature activations were extracted from the final spatial layer prior to global pooling.

- **Baseline Observation:** Unassisted models frequently activated on near-field acoustic artifacts, border tags, or bright tissue shadows.
- **CBAM Impact:** Adding spatial-channel attention constrained high activation intensity directly within the boundary margins of micro-calcifications and solid tumor masses, closely matching clinical Region of Interest (ROI) annotations.

#### 5. Future Direction & Clinical Translation

1. **Multi-Modal Cross-Attention:** Extending CBAM principles to dual-stream architectures combining ultrasound B-mode scans with strain elastography or mammography features.
2. **Edge Deployment Optimization:** Exporting PyTorch weights to TensorRT/ONNX formats to evaluate real-time inference framerates ( $> 30$  FPS) on embedded medical devices or workstation CPUs.
3. **Segmentation Integration:** Moving from image-level classification to unified segmentation-classification heads (e.g., SAM 2 / SegFormer adapters).